PhishGuard

Mobile Phishing Detection App

Team Composition

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Project Description

PhishGuard is a mobile application designed to protect users from phishing attacks by analyzing SMS messages and URLs in real-time. The application leverages machine learning algorithms combined with blacklist/whitelist databases to identify suspicious content before users interact with potentially malicious links.

The system intercepts SMS messages containing URLs, evaluates them against multiple security indicators, and provides immediate warnings when threats are detected. Users receive clear risk assessments with detailed explanations, enabling them to make informed decisions about link safety.

Motivation

Mobile phishing attacks, particularly through SMS ("smishing"), have increased by over 700% in recent years, making them one of the fastest-growing cybersecurity threats. Unlike traditional email phishing, SMS messages have higher open rates (98% vs. 20%) and users are more likely to trust them due to the personal nature of text messaging.

The average user lacks technical expertise to identify sophisticated phishing attempts that mimic legitimate organizations. Current mobile security solutions focus primarily on apps and system vulnerabilities, leaving a critical gap in SMS-based threat protection. PhishGuard addresses this urgent need by providing accessible, automated protection directly on users' devices.

Goals

The project aims to achieve the following objectives:

- **Suspicious URL Detection:** Implement pattern recognition algorithms to identify malicious URLs using domain analysis, character anomalies, and known phishing indicators
- **SMS Message Analysis:** Develop natural language processing capabilities to detect phishing indicators in message content, including urgency cues, impersonation attempts, and social engineering tactics
- **Real-time User Protection:** Create an alert system that warns users before they open dangerous links, with clear risk levels (High, Medium, Low) and actionable recommendations
- **ML Classification System:** Train and deploy a lightweight machine learning model capable of on-device classification with high accuracy (target: 95%+ detection rate with minimal false positives)

- **Database Management:** Build and maintain a continuously updated blacklist/whitelist database synchronized with threat intelligence feeds and user reports